

SPX 0DTE Live Executor

Safety C

Brief operator reference for paper and live Interactive Brokers sessions. Defaults assume LiveConfig pilot sizing (\$500k equity, 2 contracts/tranche) and production profile p3_poststop_cooldown_120. Details: live/README.md.

1. Startup interlocks

These run before any new risk is taken. Failure exits the process.

Control	What it does
Paper / live gate	mode=live also requires allow_live=True. Default stays paper.
Eligible calendar day	Skips non-tradable SPXW / era days.
VIX session skip	No trading if same-day VIX open > 35 (or VIX missing when gated).
Baselines freshness	Gated profiles require fresh signal baselines (max age 3 days).
Single-instance lock	data/live/<date>/executor.lock - second process exits if PID alive.
KILL at startup	If data/live/KILL or session KILL exists, refuse to start.
Account overlay	IB NetLiq >= configured equity; BuyingPower >= 15% of equity. PnL limits still use static account_equity.
Live market data	mode=live forces no delayed-quote fallback - fail if OPRA/index missing.

2. Restart & book recovery

- Rebuild open spreads from fills.jsonl; cancel orphan SPXW/BAG orders.
- Fail loud if IB SPXW risk does not match the recovered book.
- Restore halt / flatten / same-side cooldowns / stop counts so a restart cannot resume selling after a halt.
- Re-arm native BUY STP stops on recovered short legs.

3. In-session risk controls

Control	Trigger / action
3x short-leg stop	2-poll confirm; limit buy then MKT. Keep long wing. Confirm IB short qty dropped.
Native STP backstop	Resting BUY STP at 3x if the process dies; verify/re-arm periodically.
Side cooldown	After a stop, block new entries on that side for 120 minutes.
PnL governor	Halt new entries at -2.25% of equity; flatten open risk at -3.25%.
NetLiq loop guard	Halt entries if NetLiq < 90% of configured equity.
Mark integrity	Missing quotes: halt (never treat as \$0). Unavailable 60s with open risk: flatten.
Stale quotes	3 consecutive polls age >20s (10s near stop): halt entries only - never flatten on stale alone.
Open-risk caps	Max 6 open contracts / 3 per side / 2 same strike.
Live stop caps	Max 2 stops per side / 4 per day (entry_risk_block_reason).
Pre-entry BP	Block entry if BuyingPower < estimated margin for the size.
Entry quote hygiene	Require live NBBO, min credit \$0.20, quote age <=5s.
FOMC / VIX sizing	No new entries after 13:30 on FOMC; VIX 25-35 can upsize x1.25 (capped).
Disconnect breaker	Halt, reconnect with backoff, re-arm STPs. Fail with open risk -> confirmed flatten.
Exception path	Unexpected error: cancel pending, flatten, audit, exit.

4. Flatten confirmation & audit

Governor / kill / reconnect-fail / mark-unavailable flattens place MKT combo closes, wait for fill (one MKT retry), and only mark closed on fill. flatten_incomplete is logged if IB residual remains. flatten_audit checks IB is flat afterward and can Slack-alert on residual lots.

End of day: defined-risk 0DTE spreads are left to cash settle (matches backtest) unless flattened earlier by a safety path.

5. Operator controls (local + Slack)

- KILL file: data/live/KILL or data/live/<date>/KILL - flatten and exit. Windows: echo. > data\live\KILL
- Slack: set SPX_SLACK_WEBHOOK_URL. Alerts on halt, flatten, disconnect, kill, stop_unconfirmed, native STP reject, watchdog, etc. No-op if unset.
- Local watchdog (same machine): .\scripts\run_live_watchdog.ps1 - watches executor.lock + heartbeat.json; Slack if PID dead or heartbeat stale with open risk. Optional -WriteKill.
- Kill / heartbeat / lock are local to the host - they do not sync across PCs. Slack is the cross-device notifier.

6. Brokerage-side (every machine)

- TWS/Gateway API on; trusted 127.0.0.1; correct paper/live port.
- Precautionary max order size and daily loss; outside RTH off for SPXW.
- OPRA + US index market data (required for live mode).
- Never run two executors against the same account at once.

7. Moving to another computer

git pull; Python deps; IB Gateway login; market data; refresh baselines; set Slack webhook; start watchdog on that host; redo TWS precautionary settings. Full checklist: live/README.md - Running on another computer.

Regenerate: `python scripts/generate_live_safeties_pdf.py -> docs/Live_Executor_Safeties.pdf`